



Artificial Neural Networks

Lecture 13

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Course Progress: Where We Are and What's Ahead



Python

- Python Fundamentals



Exploratory Data Analysis

- NumPy
- Pandas
- Data Visualization
 - Matplotlib
 - Seaborn



Machine Learning



Artificial Neural Networks

Deep Learning

- Convolutional Neural Networks (Deep Computer Vision)
- Sequence Learning
- Deep Generative AI
- Large Language Models

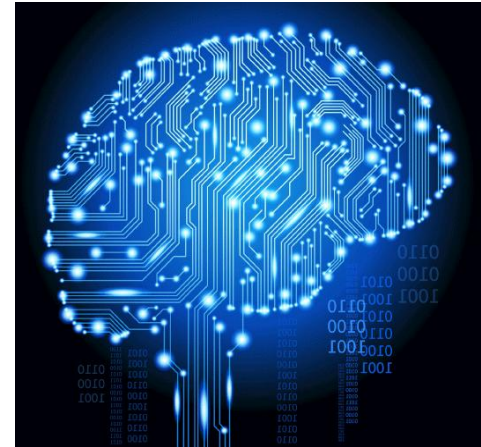
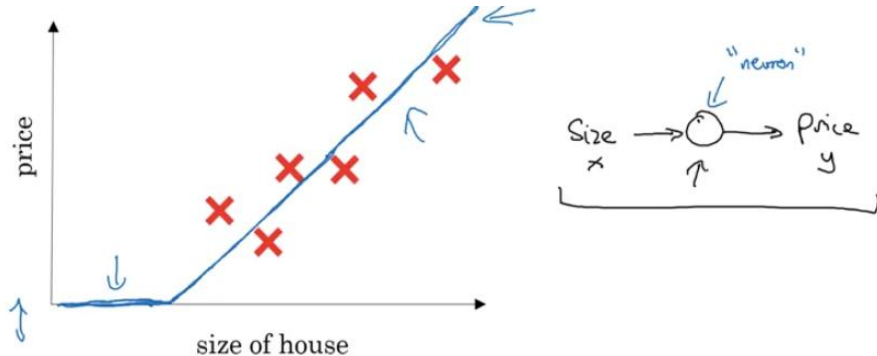
Huawei Cloud AI Services

- ModelArts (AI Development Platform)
- DAYU (Data Processing)
- AI Gallery & Pre-trained Models
- ...

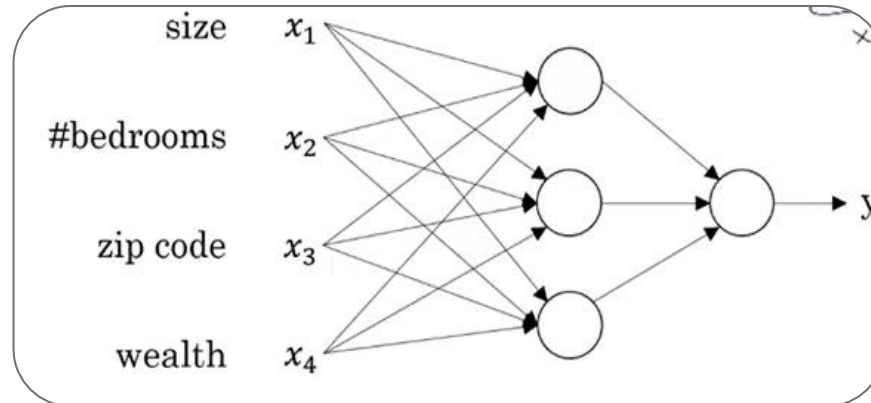
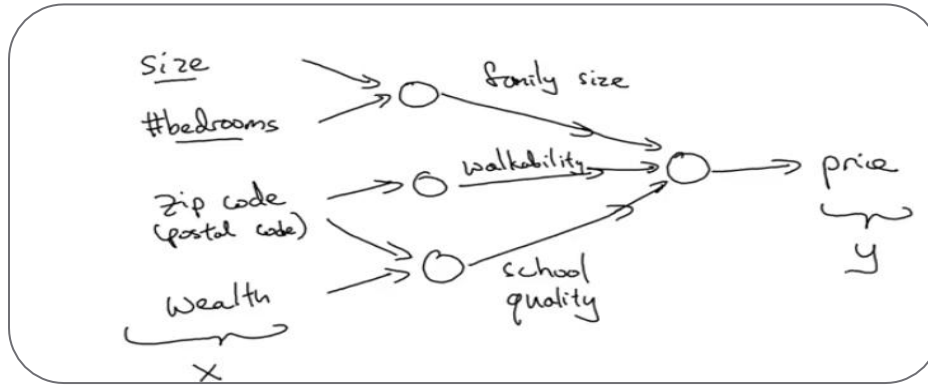


Artificial Neural Networks (ANNs)

- Neural networks are a core technique in modern AI.
- Inspired by how the human brain processes information.
- It consists of interconnected nodes (artificial neurons) that transform input data through weighted connections and activation functions.
- ANNs are used for classification, prediction, and pattern recognition across various domains.



Housing Price Prediction



Given these input features, the job of the neural network will be to predict the price y .

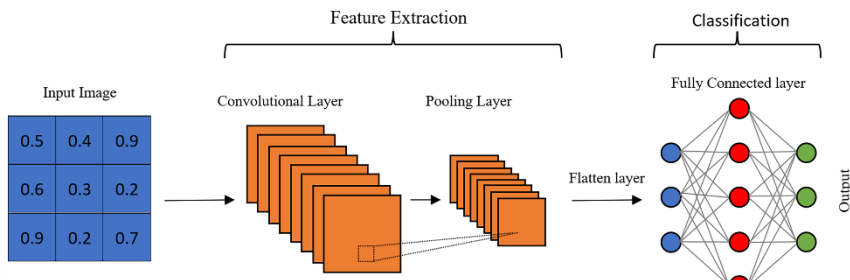
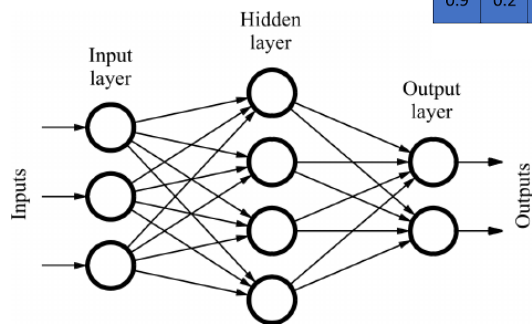
Supervised Learning with Neural Networks

Input (x)	Output (y)	Application	
Home features	Price	Real Estate	} Standard NN
Image	Objects (1, ..., 1000)	Photo tagging	
Audio	Text Transcript	Speech recognition	} CNN
English	Chinese	Machine translation	
Image, Radar Info	Position of other cars	Autonomous driving	} RNN
			} Custom

Neural Network Examples

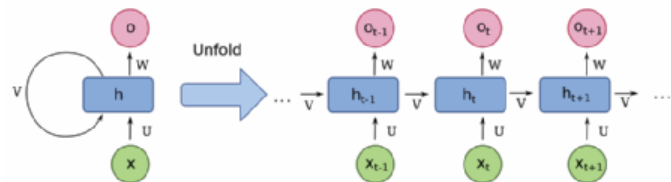
Type	Description	Application
Feedforward NN	Standard fully-connected ANN	Tabular data
CNN	Convolutional Neural Network	Image recognition
RNN	Recurrent Neural Network	Time series, text, speech

Feedforward NN



CNN

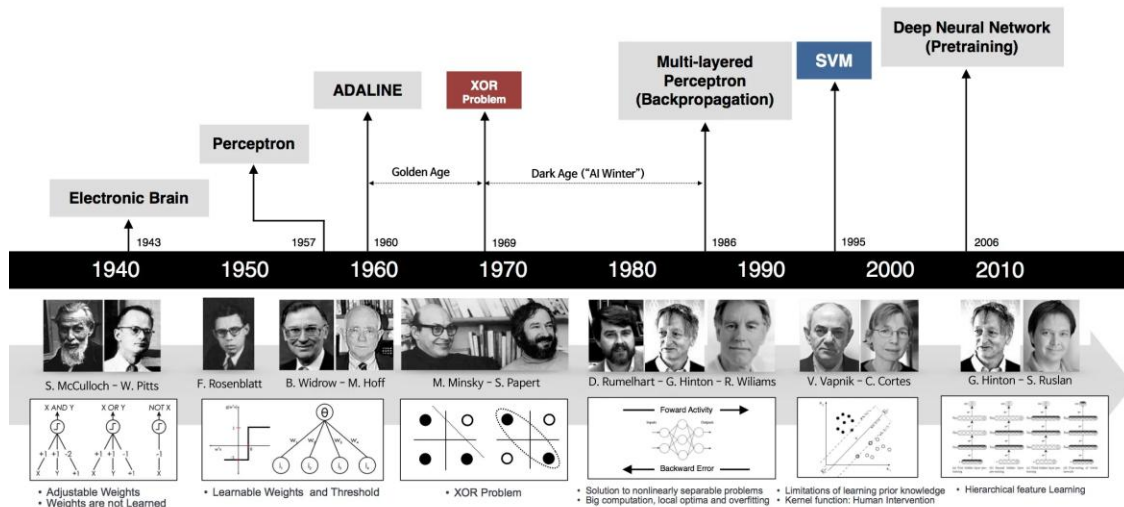
Recurrent neural network



Evolution of Neural Networks

Origins:

- Inspired by the structure of the human brain.
- Early research dates back to 1943 (McCulloch & Pitts models).
- Gained popularity in the 1980s and early 1990s.
- Interest declined in the late 1990s due to computational limitations and the rise of simpler model like Support Vector Machines (SVMs).

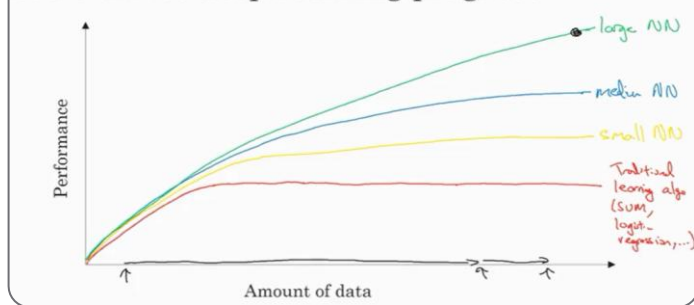


Recent Resurgence of Neural Networks

Why Now?

- Around **2005**, interest in neural networks surged again due to:
 - **Improved hardware** (GPUs, TPUs for parallel processing).
 - **Large datasets** (Big Data, Internet-scale data availability).
 - **Algorithmic advancements** (better activation functions, optimization techniques like Adam).

Scale drives deep learning progress

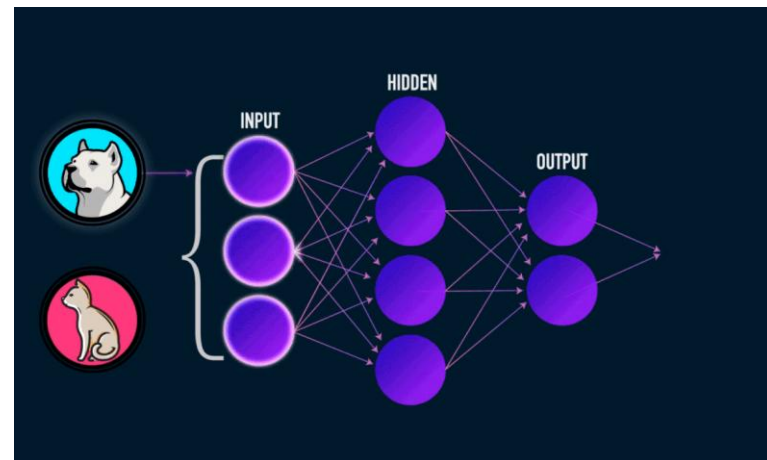
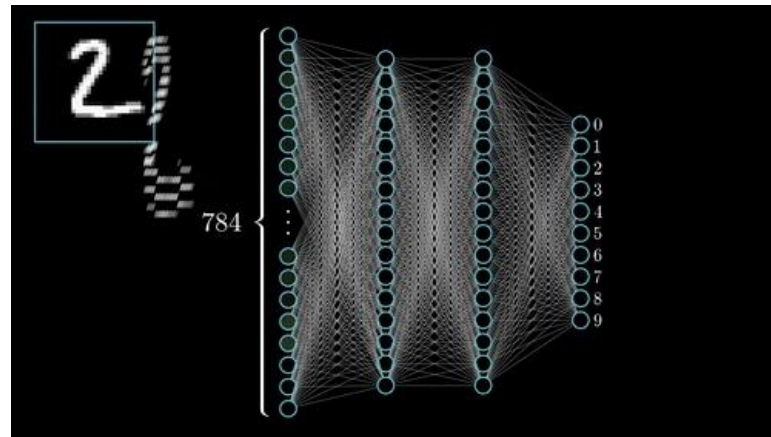


Why Neural Networks Matter?

- Neural networks power state-of-the-art applications in:
 - **Speech Recognition** (e.g., Siri, Google Assistant)
 - **Computer Vision** (e.g., Face Recognition, Object Detection)
 - **Natural Language Processing (NLP)** (e.g., ChatGPT, Google Translate)
- With enough data and computational power, they often outperform traditional machine learning.

Neural Networks – How They Work

- Neural networks learn patterns by adjusting weights of connections between neurons.
- Learning is done through backpropagation, which minimizes prediction errors using gradient descent.
- Deep networks, known as Deep Neural Networks (DNNs), consist of multiple hidden layers for hierarchical feature learning.



Neural Networks: Representation

Neurons and the brain

Biological Inspiration: The Human Brain

Biological Neurons:

- Humans have ~86 billion neurons.
- Neurons communicate via electrical and chemical signals.
- Each neuron may connect to thousands of others.
- Information processing occurs through complex networks.

From Biology to Computation:

- Artificial neurons are simplified mathematical models.
- They capture the essence of biological neural communication:
 - Receive inputs (dendrites)
 - Process signals (cell body)
 - Transmit output (axon)
- Neurons receive, process, and send signals
- Artificial neurons mimic this behavior using simple mathematical functions
- While vastly simplified, this model has proven remarkably effective

<https://articles.adxy.in>

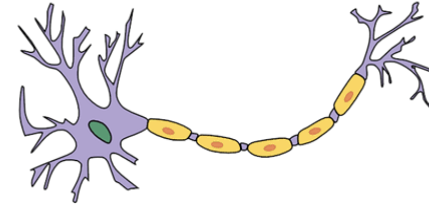


Fig: Biological Neuron

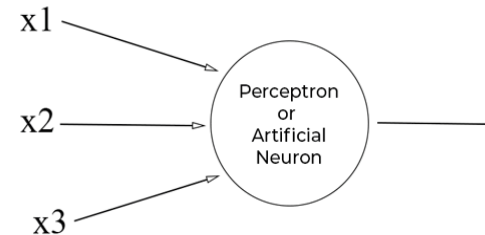
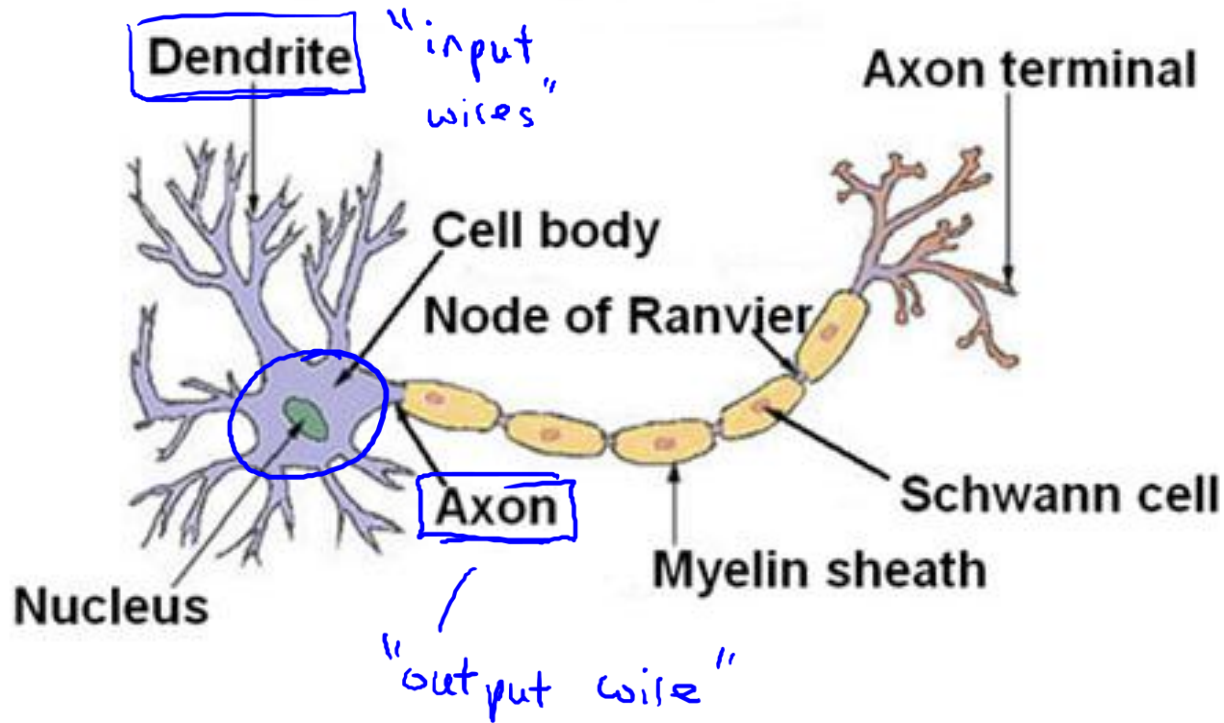
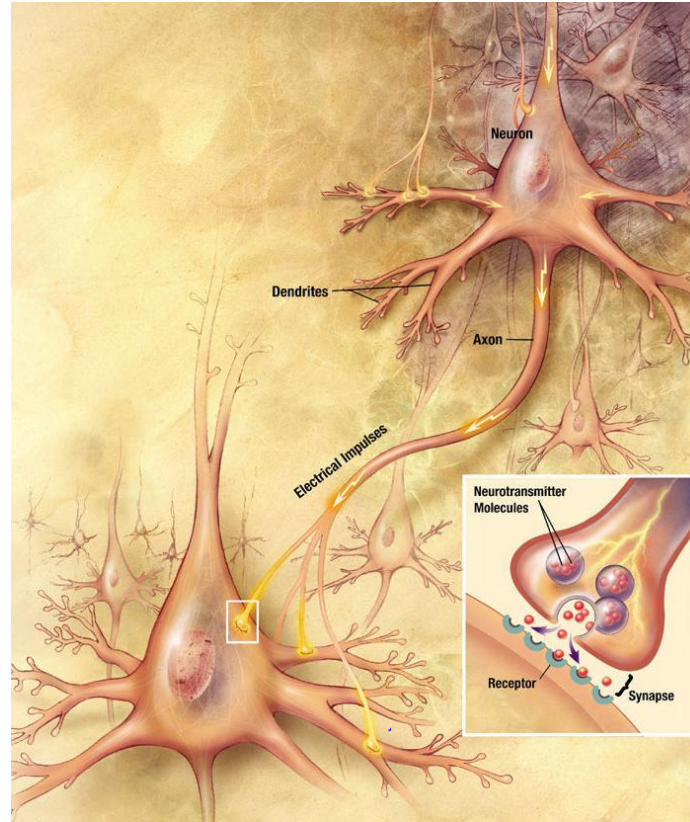


Fig: Artificial Neuron

Neuron in the brain

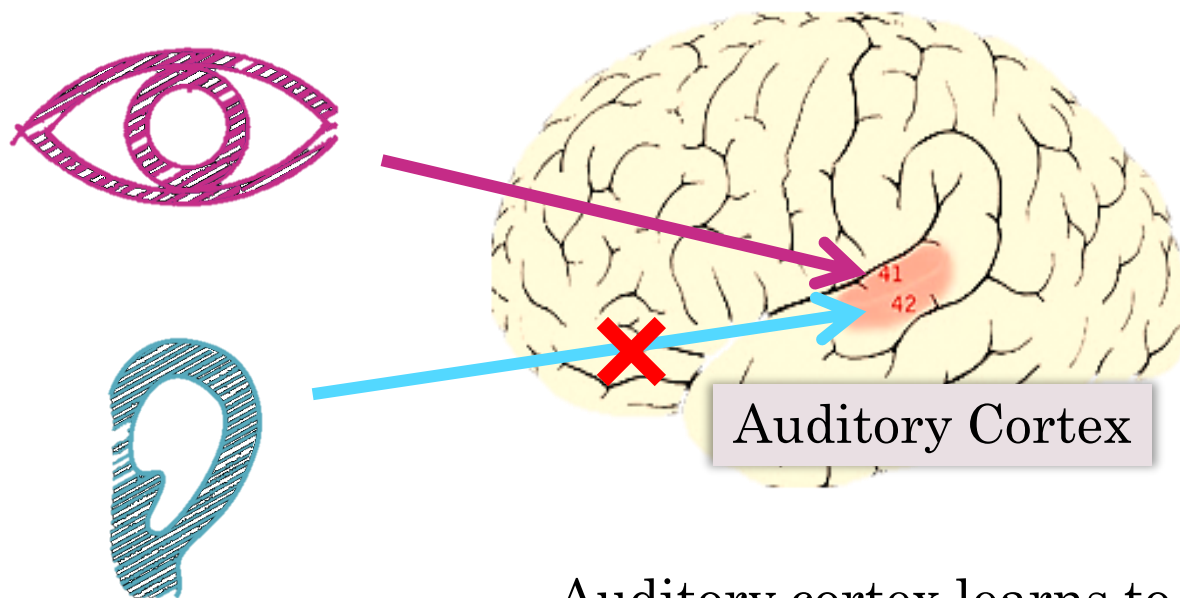


Neurons in the brain



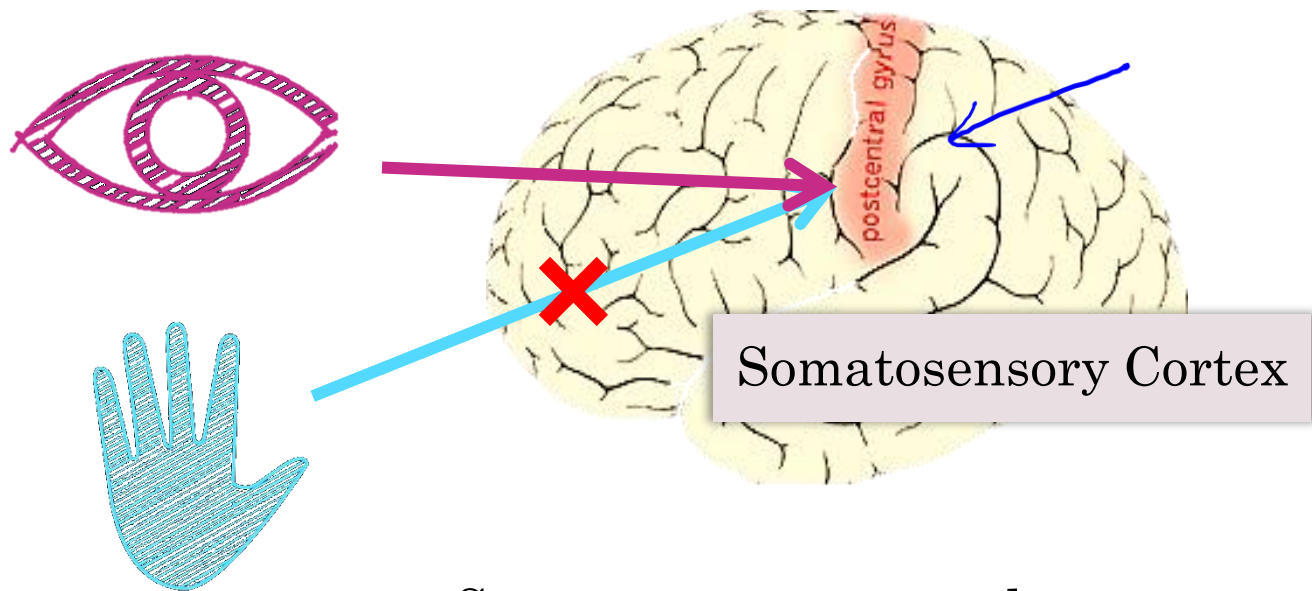
The “one learning algorithm” hypothesis

- This hypothesis suggests that the human brain does not operate with thousands of specialized algorithms.
- Instead, it relies on a single general learning mechanism to perform different tasks.
- Artificial neural networks aim to replicate this capability in a computational model.



Auditory cortex learns to see

The “one learning algorithm” hypothesis



Somatosensory cortex learns to see

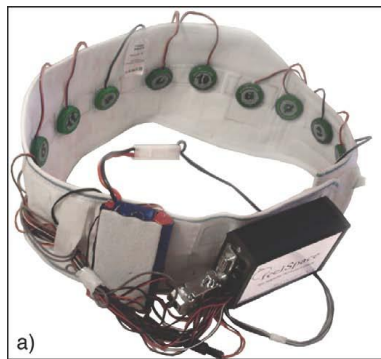
Sensor representations in the brain



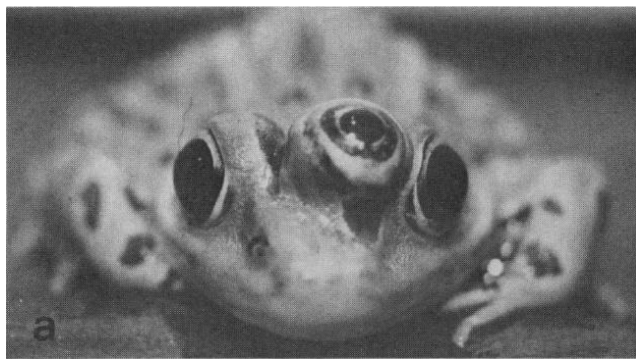
Seeing with your tongue



Human echolocation (sonar)



Haptic belt: Direction sense



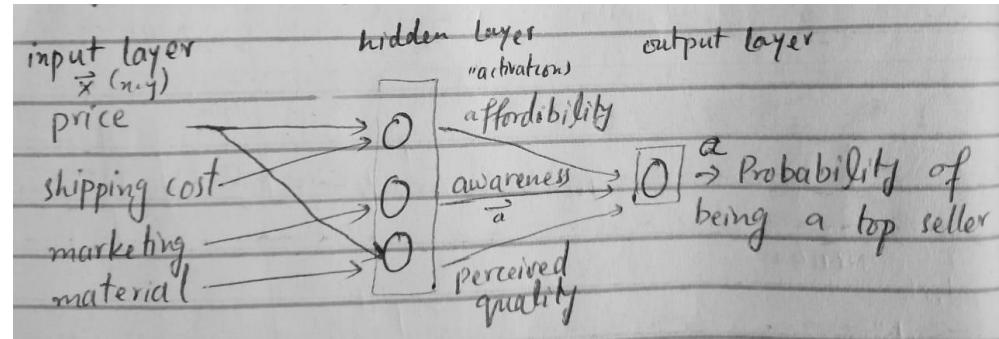
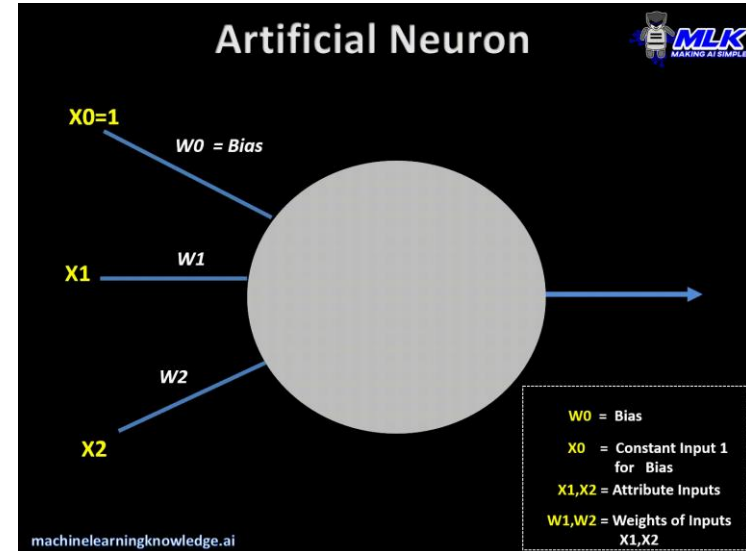
Implanting a 3rd eye

Neural Networks: Representation

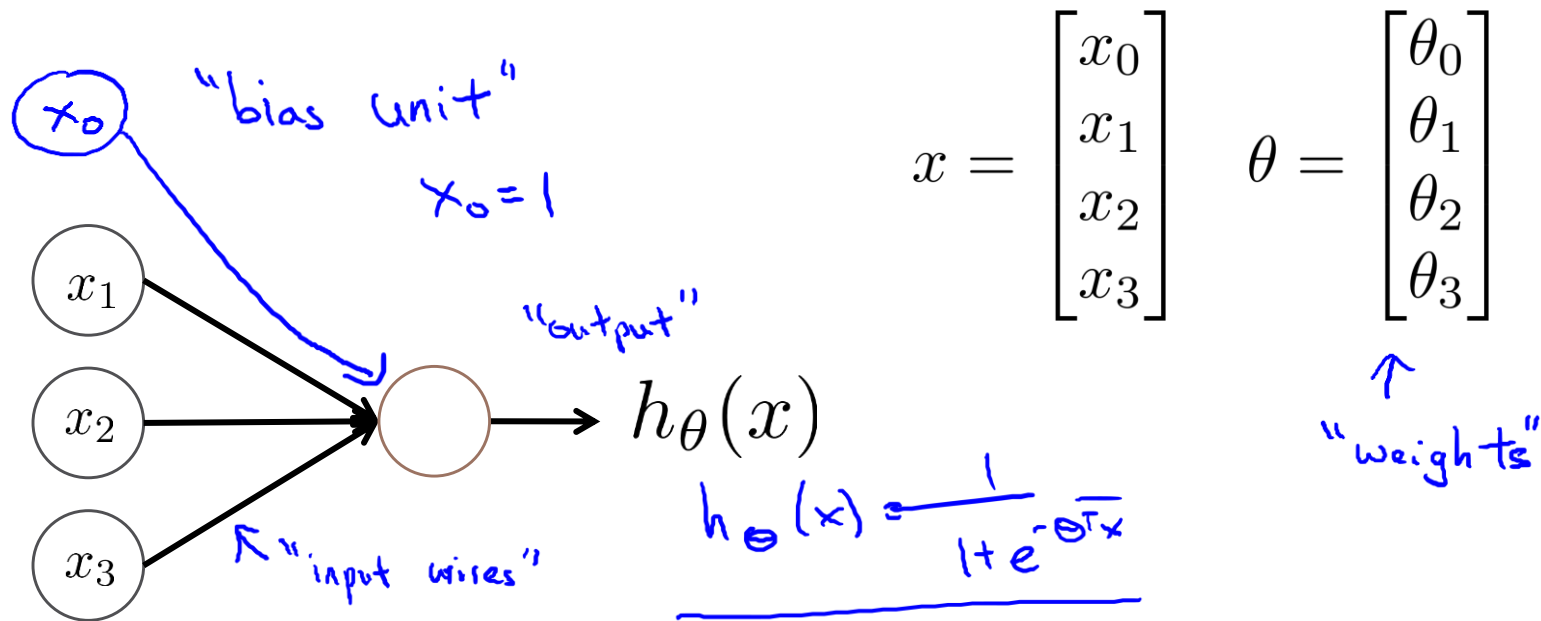
Model Representation I

Neural Network Architecture: Layers and Connections

- A neuron in an artificial neural network is a simplified model compared to biological neurons.
- A neural network consists of:
 - **Input Layer:** Takes raw data (features) as input.
 - **Hidden Layers:** Apply transformations to learn representations.
 - **Output Layer:** Produces final predictions (classification, regression, etc.).
- **Fully connected networks (Dense Networks):**
 - Each neuron in a layer connects to all neurons in the next layer.



Neuron model: Logistic unit



Sigmoid (logistic) activation function.

$$g(z) = \frac{1}{1 + e^{-z}}$$

Activation Functions

- Activation functions introduce non-linearity into the neural network, enabling it to learn and model complex patterns beyond linear relationships.

- **Common activation functions:**

- **Sigmoid:**

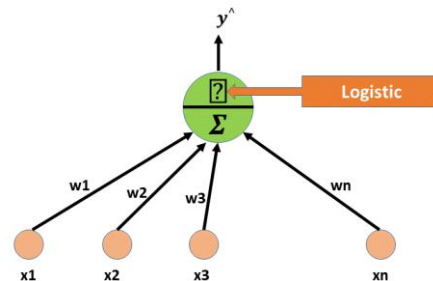
- **Output range:** $(0, 1)$
 - Good for probability based outputs (e.g., binary classification)
 - Can suffer from vanishing gradients, especially in deep networks

- **Tanh (Hyperbolic Tangent):**

- **Output range:** $(-1, 1)$
 - Centered around zero $\rightarrow$ better for optimization than sigmoid
 - Still prone to vanishing gradient issues.

- **ReLU: (Rectified Linear Unit):**

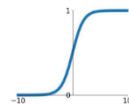
- **Output:** $\max(0, x)$
 - Most widely used due to computational efficiency
 - Helps reduce vanishing gradient problem
 - Can lead to “dead neurons” (zero gradients) if inputs are always negative.



Activation Functions

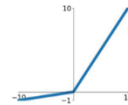
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



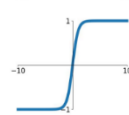
Leaky ReLU

$$\max(0.1x, x)$$



tanh

$$\tanh(x)$$

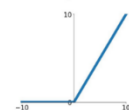


Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

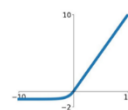
ReLU

$$\max(0, x)$$

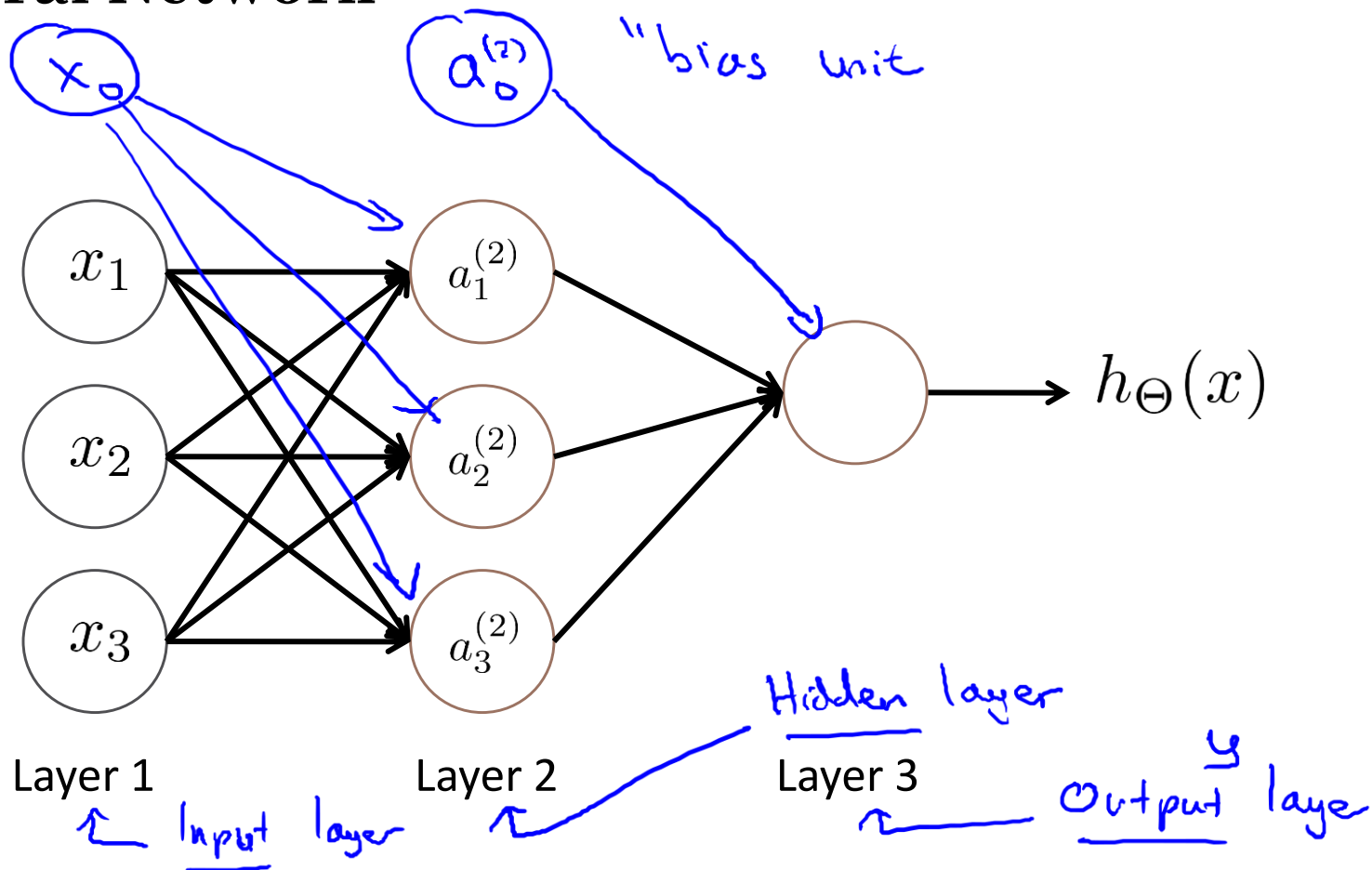


ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$

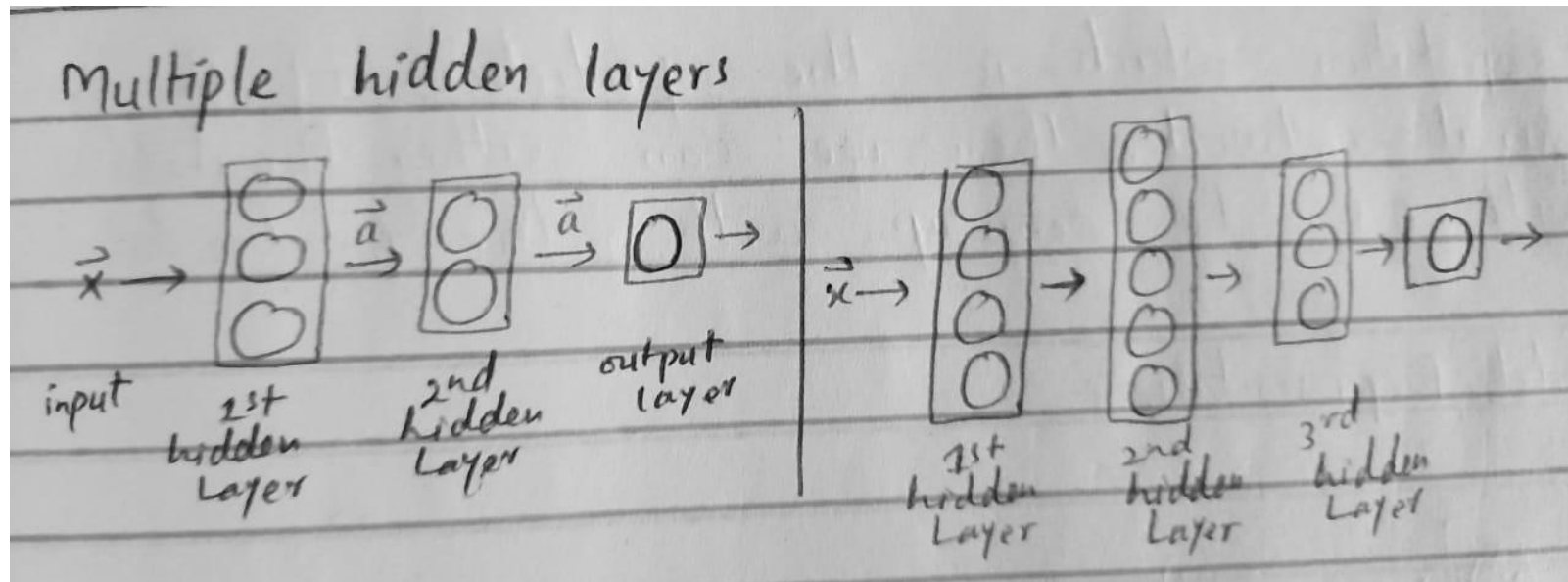


Neural Network

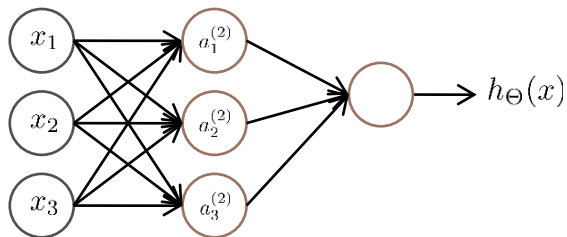


Multiple Hidden Layers

- Every layer in a neural network applies a transformation to its inputs.
- Deeper networks extract **hierarchical features**, enabling:
 - Early layers to learn **simple features** (edges, textures).
 - Deeper layers to capture **complex patterns** (objects, faces, words).
- The final output layer predicts the **desired outcome**.



Neural Network



$a_i^{(j)}$ = “activation” of unit i in layer j

$\Theta^{(j)}$ = matrix of weights controlling function mapping from layer j to layer $j + 1$

Computational Steps:

$$a_1^{(2)} = g(\Theta_{10}^{(1)} x_0 + \Theta_{11}^{(1)} x_1 + \Theta_{12}^{(1)} x_2 + \Theta_{13}^{(1)} x_3)$$

$$a_2^{(2)} = g(\Theta_{20}^{(1)} x_0 + \Theta_{21}^{(1)} x_1 + \Theta_{22}^{(1)} x_2 + \Theta_{23}^{(1)} x_3)$$

$$a_3^{(2)} = g(\Theta_{30}^{(1)} x_0 + \Theta_{31}^{(1)} x_1 + \Theta_{32}^{(1)} x_2 + \Theta_{33}^{(1)} x_3)$$

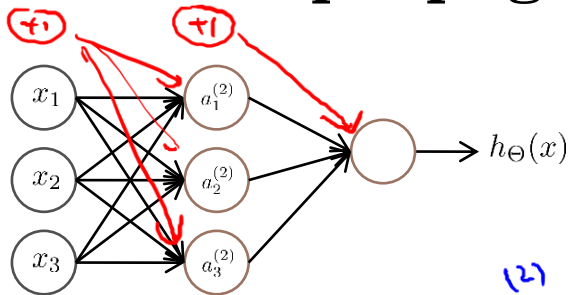
$$h_{\Theta}(x) = a_1^{(3)} = g(\Theta_{10}^{(2)} a_0^{(2)} + \Theta_{11}^{(2)} a_1^{(2)} + \Theta_{12}^{(2)} a_2^{(2)} + \Theta_{13}^{(2)} a_3^{(2)})$$

If network has s_j units in layer j , s_{j+1} units in layer $j + 1$, then $\Theta^{(j)}$ will be of dimension $s_{j+1} \times (s_j + 1)$.

Neural Networks: Representation

Model Representation II

Forward propagation: Vectorized implementation



$$a_1^{(2)} = g(z_1^{(2)})$$

$$a_1^{(2)} = g(\Theta_{10}^{(1)} x_0 + \Theta_{11}^{(1)} x_1 + \Theta_{12}^{(1)} x_2 + \Theta_{13}^{(1)} x_3)$$

$$a_2^{(2)} = g(\Theta_{20}^{(1)} x_0 + \Theta_{21}^{(1)} x_1 + \Theta_{22}^{(1)} x_2 + \Theta_{23}^{(1)} x_3)$$

$$a_3^{(2)} = g(\Theta_{30}^{(1)} x_0 + \Theta_{31}^{(1)} x_1 + \Theta_{32}^{(1)} x_2 + \Theta_{33}^{(1)} x_3)$$

$$h_{\Theta}(x) = g(\Theta_{10}^{(2)} a_0^{(2)} + \Theta_{11}^{(2)} a_1^{(2)} + \Theta_{12}^{(2)} a_2^{(2)} + \Theta_{13}^{(2)} a_3^{(2)})$$

$$a_3^{(2)} = g(z_3^{(2)})$$

$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad z^{(2)} = \begin{bmatrix} z_1^{(2)} \\ z_2^{(2)} \\ z_3^{(2)} \end{bmatrix}$$

$$z^{(2)} = \Theta^{(1)} x$$

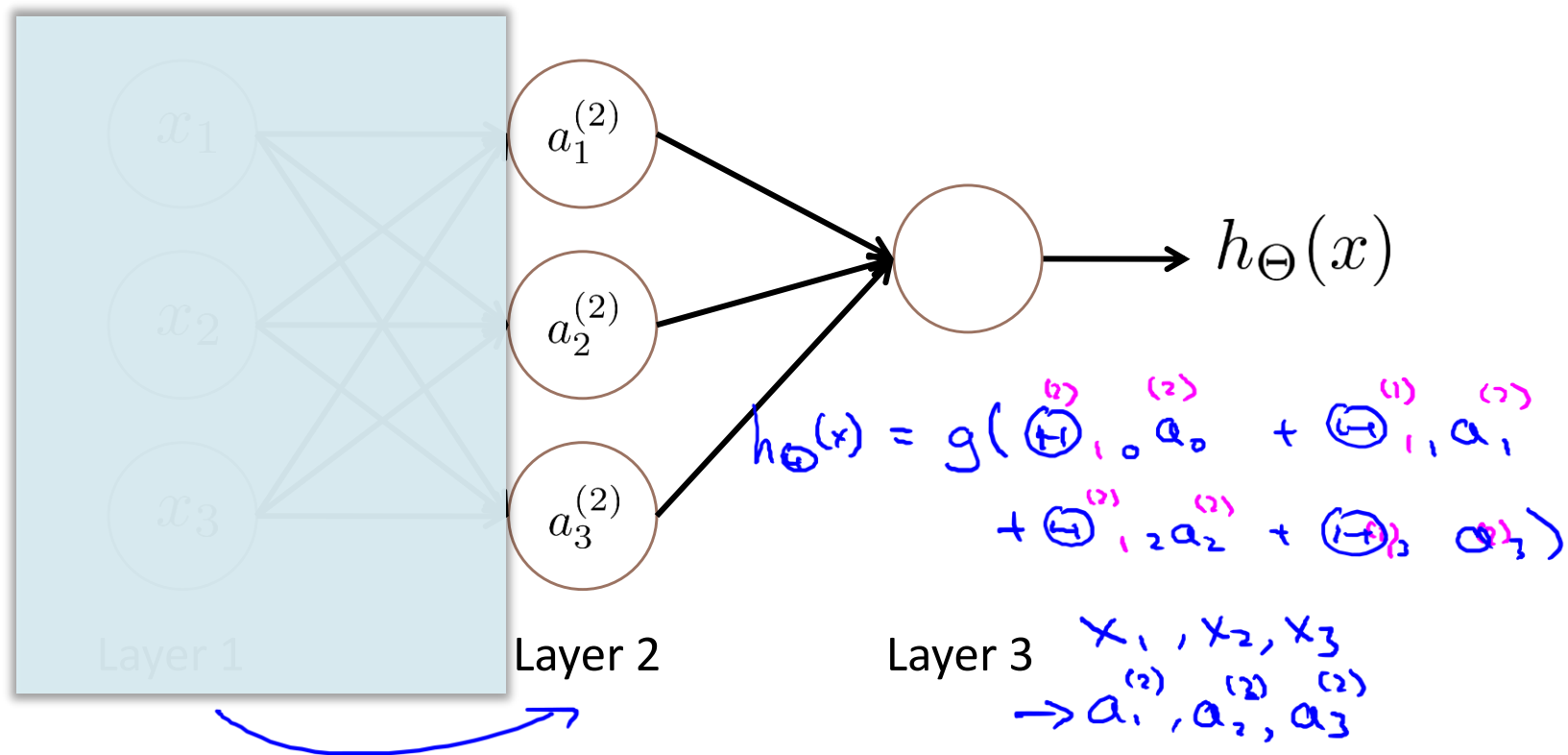
$$a^{(2)} = g(z^{(2)})$$

$$\text{Add } a_0^{(2)} = 1.$$

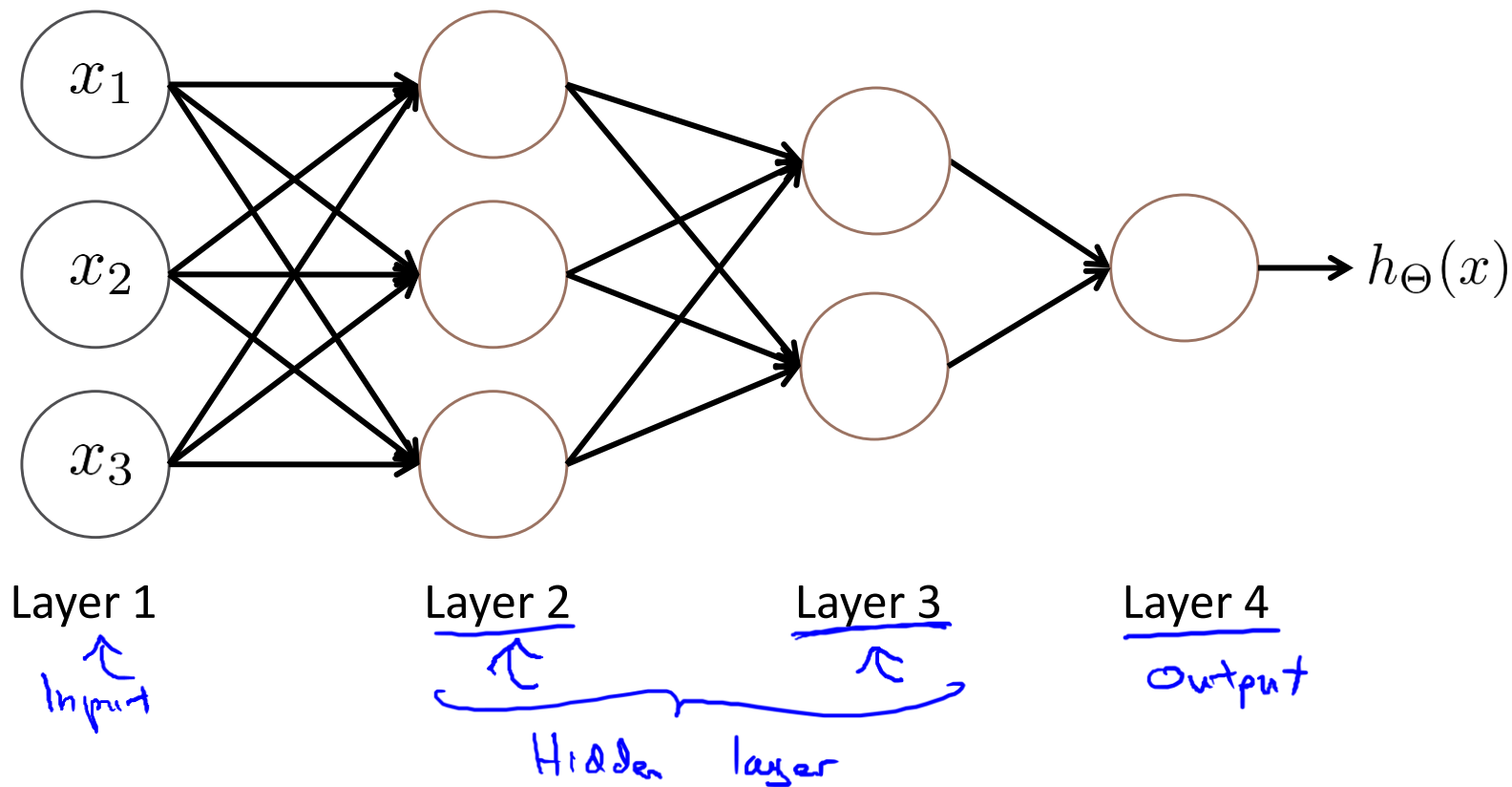
$$z^{(3)} = \Theta^{(2)} a^{(2)}$$

$$h_{\Theta}(x) = a^{(3)} = g(z^{(3)})$$

Neural Network learning its own features



Neural Network Architecture



How Neural Networks Learn

- Neural networks adjust weights and biases using labeled training data.
1. **Forward Propagation:** Compute predictions
 2. **Loss Function:** Measure error
 3. **Backpropagation:** Compute gradients
 4. **Optimization:** Update weights using gradient descent

Multi-Class Classification with ANNs



Pedestrian



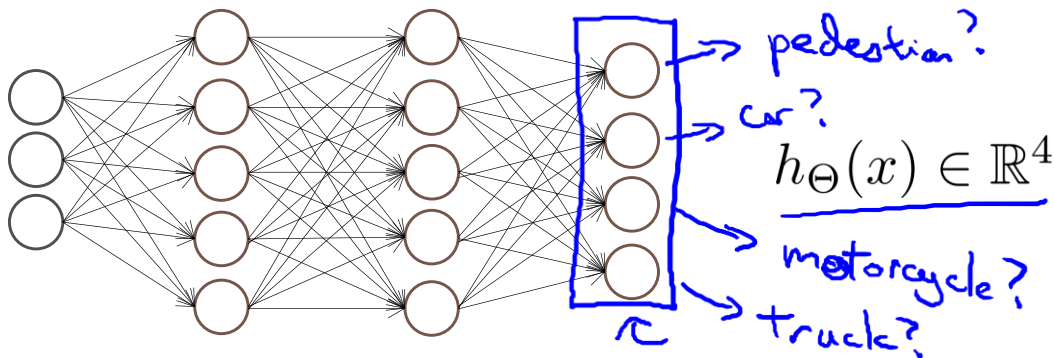
Car



Motorcycle



Truck



Our neural network output a vector of four numbers

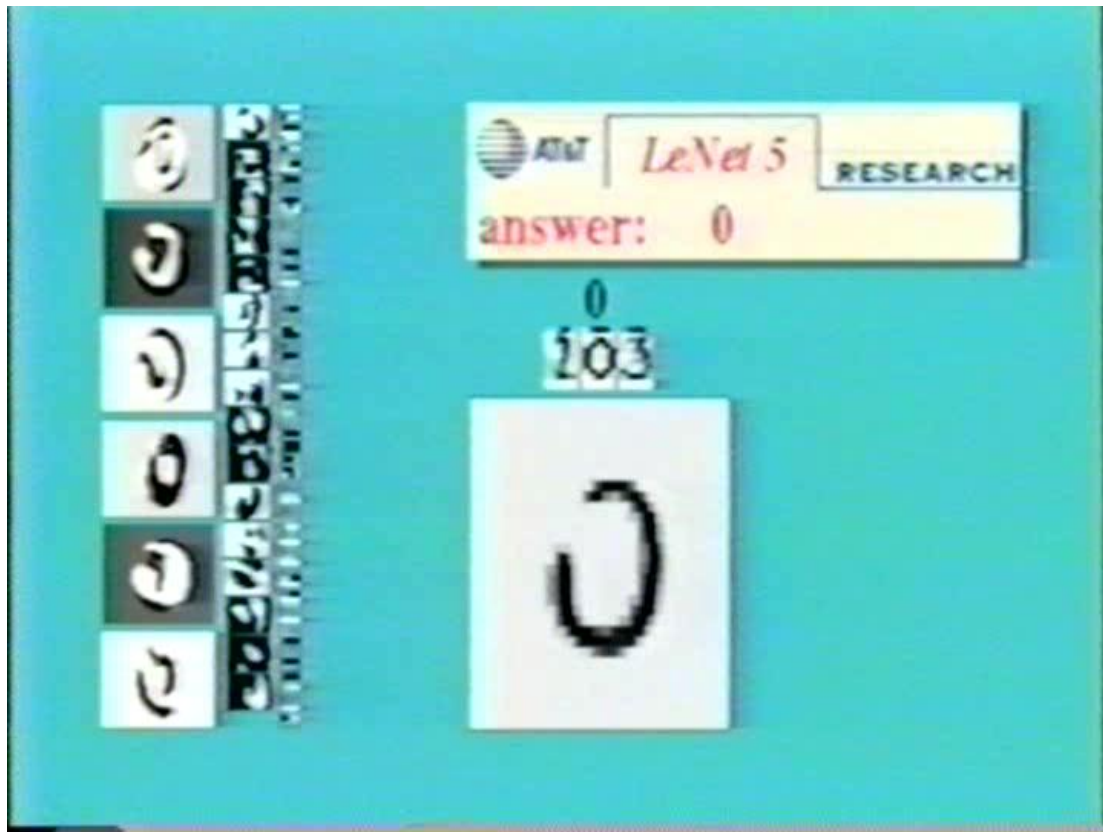
$$\text{Want } h_{\Theta}(x) \approx \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad h_{\Theta}(x) \approx \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad h_{\Theta}(x) \approx \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \text{etc.}$$

when pedestrian

when car

when motorcycle

Handwritten digit classification

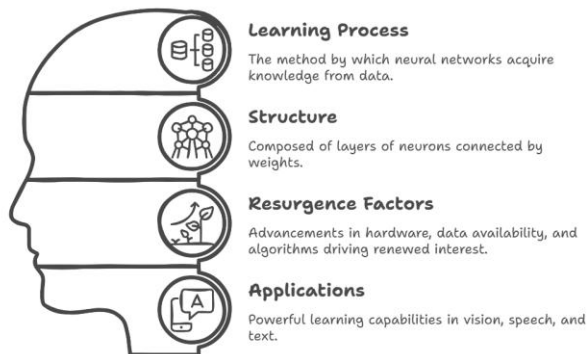


Summary

Neural Networks:

- Are inspired by the human brain
- Learn patterns from data through weights and updates.
- Work well on complex tasks like image and speech recognition
- Benefit from deeper architectures (DNNs)
- Require understanding of activation functions, layers, and the learning process.

Understanding Neural Networks



Thank You